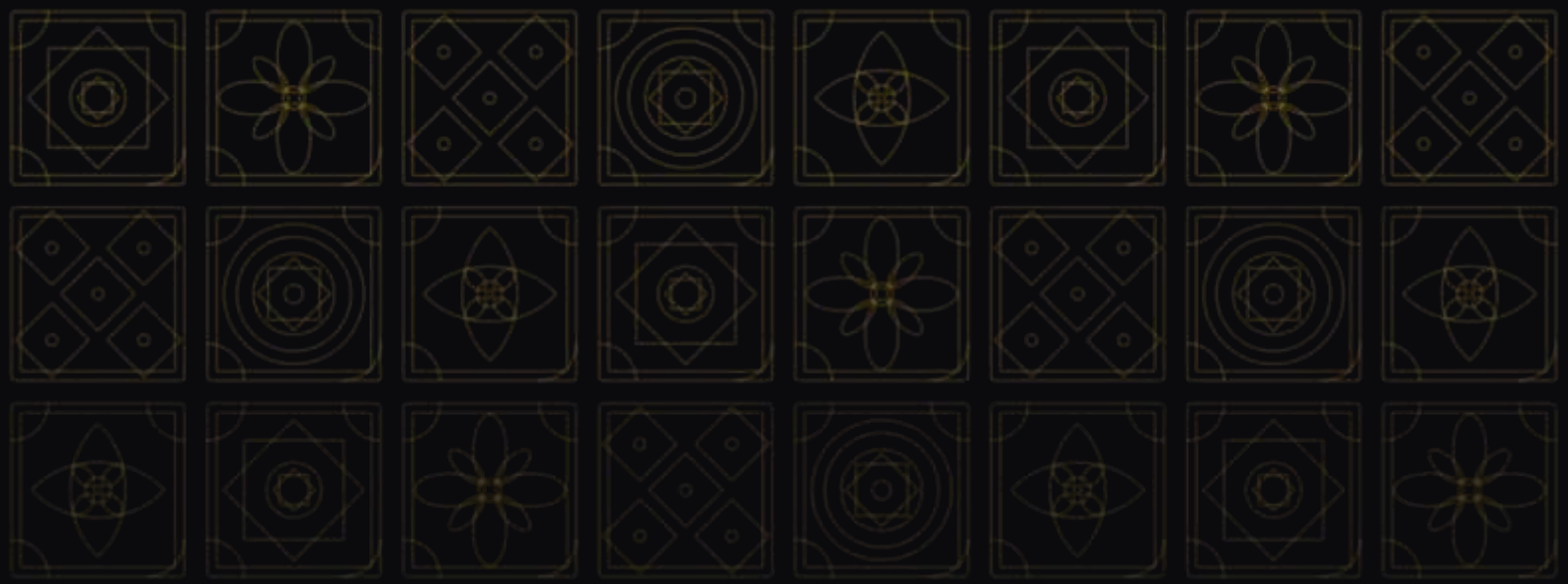




TRAINING PROGRAMME

Networking for Cloud & DevOps

The networking a developer or DevOps engineer needs to deploy and debug services

18 HOURS

FOR Developers, DevOps and cloud engineers without a networking background



Mohamed Aziz BEN HAHHA

mohamedazizbenhaha.netlify.app

Overview

This workshop gives developers and DevOps engineers the networking they meet every day when they deploy and debug services. Participants work through IP addressing and subnetting, DNS, HTTP and TLS, routing, NAT, firewalls and VPNs, then load balancers, reverse proxies and CDNs, with their cloud equivalents, always with the command-line tools used to check each layer. The workshop closes with a final project: a multi-tier web deployment with injected faults, which participants diagnose and repair.

At a glance

#	PHASE	FOCUS	HOURS
01	Network Foundations	TCP/IP, IPv4 addressing and subnetting, DNS, HTTP and TLS	6 h
02	Routing, Traffic & Troubleshooting	Routing, NAT, firewalls, load balancing and diagnostic tools	6 h
03	Final Project	Diagnose and repair a broken multi-tier deployment	3 h
	Buffer	Catch-up, revision, extra lab time	3 h
	Total		18 h

Key facts

AUDIENCE	Developers, DevOps and cloud engineers without a networking background
PREREQUISITES	None. Comfort with a Linux terminal helps (course: Linux & Bash Scripting). No networking knowledge needed
DURATION	18 h: 15 h of chapters + 3 h buffer
FORMAT	Workshop: short lectures, each followed by hands-on exercises at the command line
LAB ENVIRONMENT	Runs locally: one laptop per participant (8 GB RAM, Docker) running a multi-tier web stack in containers prepared by the trainer (DNS server, load balancer, web and database tiers); no cloud account or cost
ASSESSMENT	Exercises validated during each phase and a final troubleshooting project; certificate of completion for participants who complete both
OUTCOME	Participants can read a network path from client to service, plan subnets, and find and fix DNS, TLS, routing, firewall and load-balancer faults with standard tools

PHASE 01

6 H

Network Foundations

Participants build the mental model used for the rest of the workshop: layers, addresses and subnets, then the application protocols that every web service depends on. Validated exercise: a subnet plan for a three-tier application, and a request traced from DNS lookup to TLS handshake to HTTP response.

#	CHAPTER	CONTENT	FORMAT	HOURS
1.1	TCP/IP Essentials	OSI and TCP/IP models, Ethernet, IPv4 addressing, subnetting, CIDR	Lecture + Lab	3 h
1.2	Application Protocols	DNS, HTTP and HTTPS, TLS and certificates	Lecture + Lab	3 h

PHASE 02

6 H

Routing, Traffic & Troubleshooting

Participants follow traffic across routers, NAT and firewalls, see how load balancers, reverse proxies and CDNs spread it across servers, then practise a layer-by-layer troubleshooting method with the standard Linux tools. Validated exercise: a packet capture that shows why a request through the load balancer fails.

#	CHAPTER	CONTENT	FORMAT	HOURS
2.1	Routing & Edges	Routing, NAT, firewalls, VPN basics; cloud equivalents: VPC subnets, route tables, security groups	Lecture + Lab	1.5 h
2.2	Traffic Management	Layer 4 and layer 7 load balancing, reverse proxies, CDN; managed cloud load balancers	Lecture + Lab	1.5 h
2.3	Troubleshooting	ip, ss, dig, curl, traceroute, tcpdump	Lab	3 h

PHASE 03

3 H

Final Project

Teams of two receive a multi-tier web deployment in which the trainer has injected faults, find each fault with the tools from the workshop, fix it and explain it to the group.

#	CHAPTER	CONTENT	FORMAT	HOURS
3.1	Brief & Diagnosis	Read the incident report, map the deployment, locate the faults layer by layer	Project	1.5 h
3.2	Fix & Debrief	Fix the DNS, TLS, load-balancer and firewall faults, write the incident notes, present root causes to the group	Project	1.5 h

FINAL PROJECT

Broken Deployment Rescue

A multi-tier web deployment that does not answer: participants trace the path from client to database, find the injected DNS, TLS, load-balancer and firewall faults, and restore the service.

STACK Linux, Docker, CoreDNS, HAProxy, Nginx, OpenSSL, ip, ss, dig, curl, traceroute, tcpdump

DELIVERABLE Working deployment, incident notes with root cause for each fault